

DECLARATION

Parthjit singh [U25UV24T043076], student of 5th semester, Bachelor of Technology in Electronics and Communication Engineering, University Of Visvesvaraya College Of Engineering, KR Circle, hereby declare that the internship project titled "**AI BASED PROPULSION TEST ANALYSIS** " has been carried out by me as ENGINE TESTING ANALYSIS part of my internship work.

I further declare that the work presented in this dissertation is original to the best of my knowledge and has been completed under the prescribed academic and institutional guidelines.

I also declare that the contents of this dissertation have not been submitted previously by me or, to the best of my knowledge, by any other person for the award of any degree, diploma, or similar academic qualification at any other university or institution.

Furthermore, I declare that any Intellectual Property Rights generated from this project carried out at **CEMILAC**, Bengaluru, shall be the property of **DRDO (CEMILAC)**, Bengaluru. I acknowledge that I am the author of the submitted work and that the project outcomes, documents, and related intellectual contributions shall be governed by the applicable rules and policies of **DRDO (CEMILAC)**.

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AN INTERNSHIP REPORT

Internship Organization: DRDO, CEMILAC (Center for military Airworthiness & Certification)

Submitted By: Parthjit Singh [U25UV24T043076]

Electronics and Communication Engineering,
University Of Visvesvaraya College Of Engineering,
KR circle, Bengaluru.

Guided By: Shri. Sanyasi Rao Rupiti, Scientist 'E'

Internship Duration: 19 MAY 2026 – 06 JULY 2026

Date of Submission: 06 July 2026

SUBMITTED BY:

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the Centre for Military Airworthiness and Certification (**CEMILAC**), **DRDO**, for providing me with the valuable opportunity to undertake this project titled " **AI-Based Propulsion Test Analysis Platform**". This project has been a highly enriching learning experience and has helped me gain practical knowledge in the areas of engines, image analysis, metrics intelligence, artificial intelligence, change detection, threat assessment, and decision-support system development.

I am deeply grateful to my project guide, **Shri Sanyasi Rao Rupiti, Scientist 'E', RCMA(APS) CEMILAC** for his constant support, valuable guidance, encouragement, and technical advice throughout the project work. His mentorship played a significant role in helping me understand the practical aspects of AI-based geospatial analysis, satellite-based monitoring, and the development of an intelligent decision-support platform for identifying and assessing changes in geospatial imagery.

I would like to thank all the officials and staff members of **DRDO (CEMILAC)** for their cooperation, assistance, and the knowledge they shared with me during the course of this project. Their support and professional guidance helped me gain valuable exposure to real-world defence-oriented applications of artificial intelligence and geospatial technologies.

This project has not only enriched my academic understanding but has also enhanced my confidence in applying theoretical knowledge to practical and real-world problem domains. I am truly thankful for the exposure and learning experience received at **CEMILAC**, which will serve as a strong foundation for my future academic and professional career.

AN INTERNSHIP REPORT

Internship Organization: DRDO, CEMILAC (Center for military Airworthiness & Certification)

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Internship Duration: 19 MAY 2026 – 06 JULY 2026

Date of Submission: 06 July 2026

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DECLARATION

Jeevashree S [U25UV24T043042], student of 5th semester, Bachelor of Technology in Electronics and Communication Engineering, University Of Visvesvaraya College Of Engineering, KR Circle, hereby declare that the internship project titled " **AI BASED PROPULSION TEST ANALYSIS** " has been carried out by me as ENGINE TESTING ANALYSIS part of my internship work.

I further declare that the work presented in this dissertation is original to the best of my knowledge and has been completed under the prescribed academic and institutional guidelines.

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Prepared by:

Jeevashree S

[U25UV24T043042]

Approved by:

Shri. Sanyasi Rao Rupiti

Sc,'E', **RCMA (APS), CEMILAC**

ACKNOWLEDGEMENT

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7.1 CSV UPLOAD INTERFACE.....	43
7.2 BACKEND IMPLEMENTATION.....	44
7.3 HISTORICAL TEST REPOSITORY.....	45
CHAPTER 8 – RESULTS AND PERFORMANCE EVALUATION... ..	49
8.1 INTRODUCTION	49
8.2 TESTING METHODOLOGY	49
8.3 FUNCTIONAL TESTING	50
8.4 USER INTERFACE EVALUATION.....	51
8.5 PERFORMANCE EVALUATION.....	51
8.6 RESULT ANALYSIS	52
8.7 ADVANTAGES OBSERVED DURING TESTING.....	53
CHAPTER 9 – ADVANTAGES, LIMITATIONS AND FUTURE SCOPE.....	54
9.1 ADVANTAGES OF THE PROPOSED SYSTEM... ..	54
9.2 INDUSTRIAL APPLICATIONS.....	54
9.3 ACHIEVEMENT OF PROJECT OBJECTIVES.....	55
9.4 MAJOR CONTRIBUTIONS.....	55
9.5 BENEFITS.....	55
9.6 FUTURE SCOPE... ..	55
CHAPTER 10 – REFERENCES AND BIBIOGRAPHY.....	56
10.1 INTRODUCTION	56
10.2 BOOKS.....	57
10.3 RESEARCH PAPERS.....	57
10.4 OFFICIAL DOCUMENTATION.....	57
10.5 OFFLINE LEARNING RESOURCES.....	58
10.6 WEBSITES REFERENCED	58
10.7 BIBIOGRAPHY.....	58
CHAPTER 11 – APPENDIX, WORKFLOW.....	59

7.1 CSV UPLOAD INTERFACE.....	43
7.2 BACKEND IMPLEMENTATION.....	44
7.3 HISTORICAL TEST REPOSITORY	45
CHAPTER 8 – RESULTS AND PERFORMANCE EVALUATION... ..	49
8.1 INTRODUCTION	49
8.2 TESTING METHODOLOGY	49
8.3 FUNCTIONAL TESTING	50
8.4 USER INTERFACE EVALUATION.....	51
8.5 PERFORMANCE EVALUATION.....	51
8.6 RESULT ANALYSIS	52
8.7 ADVANTAGES OBSERVED DURING TESTING.....	53
CHAPTER 9 – ADVANTAGES, LIMITATIONS AND FUTURE SCOPE.....	54
9.1 ADVANTAGES OF THE PROPOSED SYSTEM... ..	54
9.2 INDUSTRIAL APPLICATIONS.....	54
9.3 ACHIEVEMENT OF PROJECT OBJECTIVES.....	55
9.4 MAJOR CONTRIBUTIONS.....	55
9.5 BENEFITS.....	55
9.6 FUTURE SCOPE... ..	55
CHAPTER 10 – REFERENCES AND BIBILOGRAPHY	56
10.1 INTRODUCTION	56
10.2 BOOKS.....	57
10.3 RESEARCH PAPERS.....	57
10.4 OFFICIAL DOCUMENTATION.....	57
10.5 OFFLINE LEARNING RESOURCES.....	58
10.6 WEBSITES REFERENCED	58
10.7 BIBILOGRAPHY.....	58
CHAPTER 11 – APPENDIX, WORKFLOW.....	59

CHAPTER 11 – APPENDIX, WORKFLOW

drdo-propulsion-platform

```
|
|__backend
|  |__analytics_engine  (models, views, serializers, services,
admin, apps, urls,
|  |  calculations, constants, health_score,
llm_service, chat_views, tests, migrations)
|  |__chat              (models, views, admin, apps, urls, tests,
migrations)
|  |__config            (settings.py, urls.py, wsgi.py, asgi.py)
|  |__media\csv         (sample_engine_test_TEST_13011.csv,
TEST_278452.csv + variants)
|  |__rag               (chunker, embeddings, loader,
llm_service, prompt_builder,
|  |  report_prompt, report_service,
retriever, telemetry_context, vector_store, warmup, tests)
|  |__reports_engine    (models, views, admin, apps, tests,
migrations)
|  |__telemetry         (models, views, serializers, admin, apps,
urls, tests, migrations)
|  |__tests_app         (models, views, serializers, services,
admin, apps, urls, tests, migrations)
|  |__venv, db.sqlite3, manage.py
|
|__docs
|
|__frontend
|  |__public            (drdo-logo.png, favicon.svg, icons.svg)
|  |__src
|    |__components      (Assistant, Graphs, Header, Metrics,
Repository, Upload)
|    |__context          (SelectedTestContext.tsx)
|    |__pages            (Dashboard.tsx)
|    |__services         (api.ts)
|    |__styles           (dashboard.css)
|    |__types            (api.ts, chat.ts, dashboard.ts,
telemetry.ts)
|    |__App.tsx, App.css, main.tsx
|    |__node_modules
|    |__package.json, package-lock.json, tsconfig files,
vite.config.ts, index.html, README.md
|
|__rag_data
|
|__reports
|
|__uploaded_files
```

CHAPTER 11 – APPENDIX,WORKFLOW

drdo-propulsion-platform

```
|
|__backend
|  |__analytics_engine  (models, views, serializers, services,
admin, apps, urls,
|  |  calculations, constants, health_score,
llm_service, chat_views, tests, migrations)
|  |__chat              (models, views, admin, apps, urls, tests,
migrations)
|  |__config            (settings.py, urls.py, wsgi.py, asgi.py)
|  |__media\csv         (sample_engine_test_TEST_13011.csv,
TEST_278452.csv + variants)
|  |__rag               (chunker, embeddings, loader,
llm_service, prompt_builder,
|  |  report_prompt, report_service,
retriever, telemetry_context, vector_store, warmup, tests)
|  |__reports_engine    (models, views, admin, apps, tests,
migrations)
|  |__telemetry         (models, views, serializers, admin, apps,
urls, tests, migrations)
|  |__tests_app         (models, views, serializers, services,
admin, apps, urls, tests, migrations)
|  |__venv, db.sqlite3, manage.py
|
|__docs
|
|__frontend
|  |__public            (drdo-logo.png, favicon.svg, icons.svg)
|  |__src
|    |__components     (Assistant, Graphs, Header, Metrics,
Repository, Upload)
|    |__context        (SelectedTestContext.tsx)
|    |__pages          (Dashboard.tsx)
|    |__services       (api.ts)
|    |__styles         (dashboard.css)
|    |__types          (api.ts, chat.ts, dashboard.ts,
telemetry.ts)
|    |__App.tsx, App.css, main.tsx
|    |__node_modules
|    |__package.json, package-lock.json, tsconfig files,
vite.config.ts, index.html, README.md
|
|__rag_data
|
|__reports
|
|__uploaded_files
```